SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
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॥वसुधैव कुटुम्बकम्॥

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Subject: Programming with Java	**Sem: III AY: 2026-27**

Assignment No: 8

TITLE: Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

1.Create an Employee and Manager program where Manager accesses Employee details using super keyword.

Main.java

Share

Run

```java
class Employee {
    String name;
    int id;

    Employee(String name, int id) {
        this.name = name;
        this.id = id;
    }
}

class Manager extends Employee {
    String department;

    Manager(String name, int id, String department) {
        super(name, id);
        this.department = department;
    }

    void display() {
        System.out.println("Employee Name: " + super.name);
        System.out.println("Employee ID: " + super.id);
        System.out.println("Department: " + department);
    }
}

class Main {
    public static void main(String[] args) {
        Manager m = new Manager("Aryan Singh Kasera",122052, "Sales");

        m.display();
    }
```

OUTPUT

Output

```
Employee Name: Aryan Singh Kasera
Employee ID: 122052
Department: Sales

=== Code Execution Successful ===
```

2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

Main.java

```java
class Vehicle {
    String vehicleNo;
    String ownerName;

    Vehicle(String vehicleNo, String ownerName) {
        this.vehicleNo = vehicleNo;
        this.ownerName = ownerName;
    }
}

class CarInsurance extends Vehicle {
    double premium;

    CarInsurance(String vehicleNo, String ownerName, double premium) {
        super(vehicleNo, ownerName);
        this.premium = premium;
    }

    void display() {
        System.out.println("Vehicle Number: " + super.vehicleNo);
        System.out.println("Owner Name: " + super.ownerName);
        System.out.println("Insurance Premium: " + premium);
    }
}

class Main {
    public static void main(String[] args) {
        CarInsurance c = new CarInsurance("UP32AS7777", "Aryan Singh", 19000);
     c.display();
    }
}
```

OUTPUT

Output

```
Vehicle Number: UP32AS7777
Owner Name: Aryan Singh
Insurance Premium: 19000.0

=== Code Execution Successful ===
```

Conclusion:

This experiment demonstrated the use of the **super** keyword in Java inheritance. The `super` keyword was used to access parent class variables, invoke parent class methods, and call the parent class constructor, ensuring proper initialization of inherited objects. It also helped distinguish between parent and child class members with the same name. Overall, the `super` keyword plays an essential role in inheritance by enabling constructor chaining, code reuse, and effective implementation of method overriding in object-oriented programming.